

8.3 Experimental Design II: Randomized Complete Block Designs and Pseudo-replication

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Randomized complete block designs

$$\sigma^2 = \sigma_{bio}^2 + \sigma_{lab}^2 + \sigma_{extraction}^2 + \sigma_{run}^2 + \dots$$

- Biological: fluctuations in protein level between mice, fluctuations in protein level between cells, ...
- Technical: cage effect, lab effect, week effect, plasma extraction, MS-run, ...

Nature methods: Points of significance - Blocking

<https://www.nature.com/articles/nmeth.3005.pdf>

- Oneway anova is a special case of a completely randomized design:

- the experimental units are sampled at random from the population
 - the treatments are randomized to the experimental units
 - Every experimental unit receives one treatment and its response variable is measured once.
- In a block design the experimental units are blocks which are sampled at random of the population of all possible blocks.
 - The RCB restricts randomization: the treatments are randomized within the blocks.
 - it cannot be analysed using oneway anova.
 - The paired design is a special case of an RCB: block size equals 2.
 - Within block effects can be assessed using the `lm` function in R.

Mouse example

Background

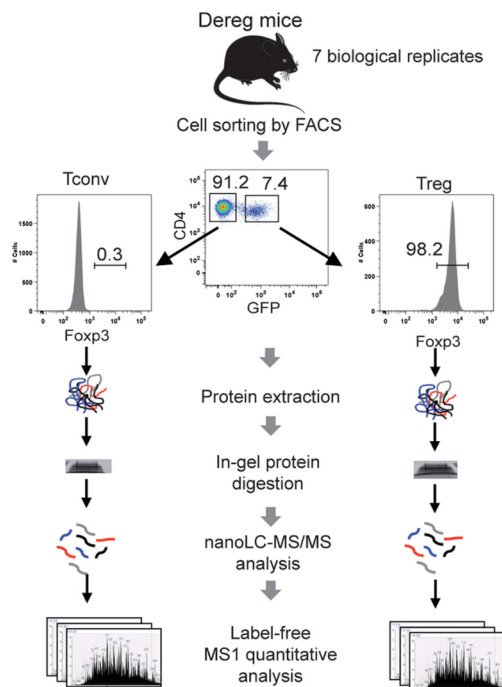


FIG. 1. **Label-free quantitative analysis of conventional and regulatory T cell proteomes.** General analytical workflow based on cell sorting by flow cytometry using the DEREG mouse model and parallel proteomic analysis of Tconv and Treg cell populations by nanoLC-MS/MS and label-free relative quantification.

Duguet et al. (2017) MCP 16(8):1416-1432. doi: 10.1074/mcp.m116.062745

- All treatments of interest are present within block!
- We can estimate the effect of the treatment within block!

We focus on one protein

- The measured intensities are already on the log2-scale. Differences between the intensities can thus be interpreted as log2 FC.
- P16045 or Galectin-1.
- Function: “Lectin that binds beta-galactoside and a wide array of complex carbohydrates. Plays a role in regulating apoptosis, cell proliferation and cell differentiation. Inhibits CD45 protein phosphatase activity and therefore the dephosphorylation of Lyn kinase. Strong inducer of T-cell apoptosis.” (source: [uniprot](#))

Data Exploration

```
mouse <- read_tsv("https://raw.githubusercontent.com/statOmics/PSLSDData/main/mouseP16045.txt")
```

```
Rows: 14 Columns: 3
```

```
-- Column specification -----
```

```
Delimiter: "\t"
```

```
chr (2): celltype, mouse
```

```
dbl (1): intensity
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

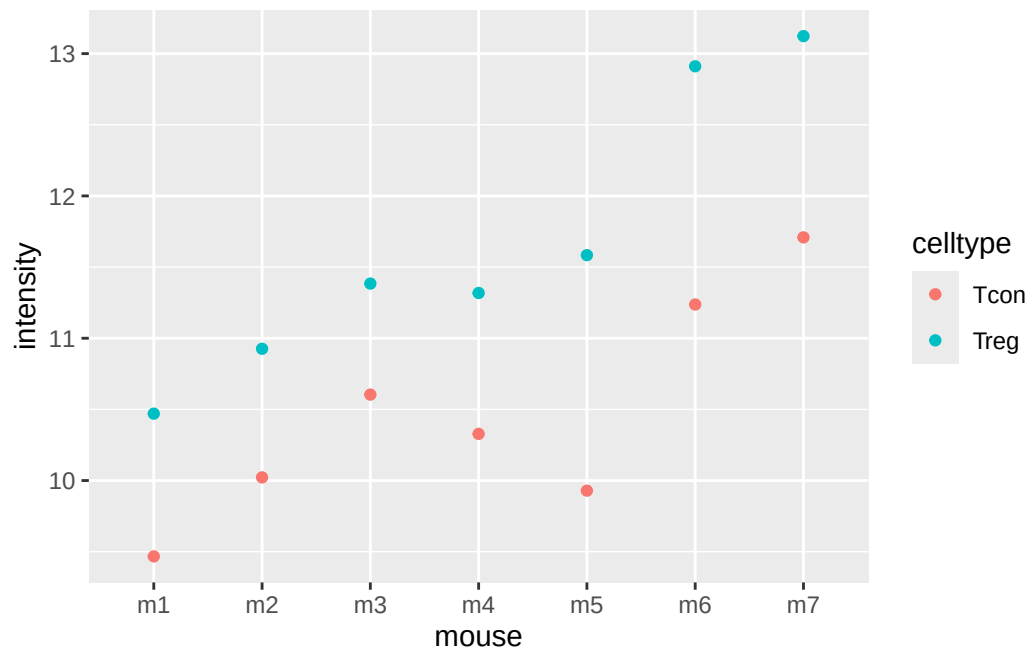
```
mouse
```

```
# A tibble: 14 x 3
```

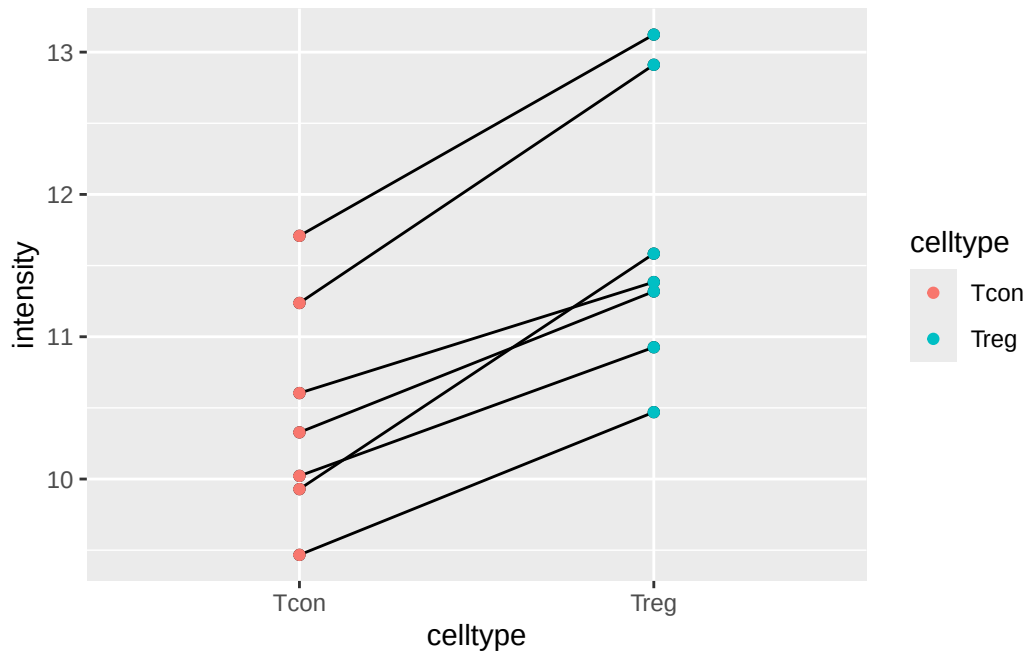
	celltype	mouse	intensity
	<chr>	<chr>	<dbl>
1	Tcon	m1	9.47
2	Tcon	m2	10.0
3	Tcon	m3	10.6
4	Tcon	m4	10.3
5	Tcon	m5	9.93
6	Tcon	m6	11.2
7	Tcon	m7	11.7
8	Treg	m1	10.5
9	Treg	m2	10.9

10	Treg	m3	11.4
11	Treg	m4	11.3
12	Treg	m5	11.6
13	Treg	m6	12.9
14	Treg	m7	13.1

```
mouse |>
  ggplot(aes(x = mouse, y = intensity, col = celltype)) +
  geom_point()
```



```
mouse |>
  ggplot(aes(x = celltype, y = intensity)) +
  geom_point() +
  geom_line(aes(group = mouse)) +
  geom_point(aes(col = celltype))
```



The plots show evidence for - an upregulation of the protein expression in regulatory T-cells and - a considerable variability in expression from animal to animal!

Paired Analysis

This is a paired design, which is the most simple form of randomized complete block design.

In the introduction to statistical inference we would have analyzed the data by differencing.

```
mouseWide <- mouse |>
  spread(celltype, intensity) |>
  mutate(delta = Treg - Tcon)
mouseWide
```

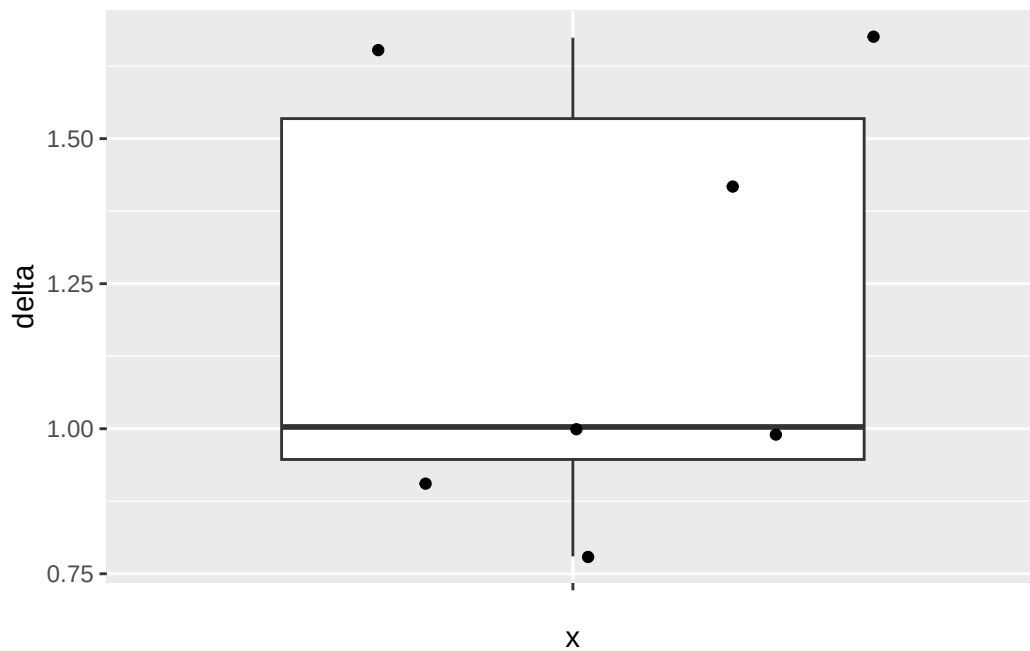
```
# A tibble: 7 x 4
  mouse Tcon  Treg delta
<chr> <dbl> <dbl> <dbl>
1 m1    9.47  10.5  1.00
2 m2   10.0  10.9  0.904
3 m3   10.6  11.4  0.780
4 m4   10.3  11.3  0.99
5 m5    9.93  11.6  1.65
6 m6   11.2  12.9  1.67
```

7 m7 11.7 13.1 1.41

Data exploration

- Boxplot of difference

```
mouseWide |>
  ggplot(aes(x = "", y = delta)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter()
```



- Summary statistics

```
deltaSum <- mouseWide |>
  summarize(
    mean = mean(delta, na.rm = TRUE),
    sd = sd(delta, na.rm = TRUE),
    n = n()
  ) |>
  mutate(se = sd / sqrt(n))
deltaSum
```

```
# A tibble: 1 x 4
  mean    sd    n    se
<dbl> <dbl> <int> <dbl>
1  1.20 0.371     7 0.140
```

Note, that the intensity data are not independent because we measured the expression in regulatory and ordinary T-cells for every animal

- Covariance and correlation between expression in both celltypes

```
cor(mouseWide[, c("Tcon", "Treg")])
```

```
      Tcon    Treg
Tcon 1.00000 0.93874
Treg 0.93874 1.00000
```

```
var(mouseWide[, c("Tcon", "Treg")])
```

```
      Tcon    Treg
Tcon 0.6101531 0.7245316
Treg 0.7245316 0.9763042
```

```
var(mouseWide[, c("Tcon", "Treg")]) |>
  diag() |>
  sqrt()
```

```
      Tcon    Treg
0.7811230 0.9880811
```

- There is indeed a large correlation between the expression of the protein in conventional and regulatory T-cells.
- Standard deviation of difference?

$$\begin{aligned} \text{sd}_{x_r - x_c} &= \sqrt{1^2 \hat{\sigma}_r^2 + (-1)^2 \hat{\sigma}_c^2 + 2 \times 1 \times -1 \times \hat{\sigma}_{r,c}} \\ &= \sqrt{\hat{\sigma}_r^2 + \hat{\sigma}_c^2 - 2 \times \hat{\sigma}_{r,c}} \end{aligned}$$

```
sdDelta2 <- (c(-1, 1) %*% var(mouseWide[, c("Tcon", "Treg")]) %*% c(-1, 1)) |>
  sqrt()
sdDelta2
```

```
      [,1]  
[1,] 0.3706672
```

```
seDeltaBar <- sdDelta2 / sqrt(deltaSum$n)  
seDeltaBar
```

```
      [,1]  
[1,] 0.140099
```

```
deltaSum
```

```
# A tibble: 1 x 4  
  mean    sd    n    se  
  <dbl> <dbl> <int> <dbl>  
1  1.20 0.371     7 0.140
```

- The standard deviation on the difference is much lower because of the strong correlation!
- Note, that the paired design enabled us to estimate the difference in log2-expression between the two cell types in every animal (log2 FC).

```
t.test(delta~1, mouseWide)
```

One Sample t-test

```
data: delta  
t = 8.5858, df = 6, p-value = 0.0001372  
alternative hypothesis: true mean is not equal to 0  
95 percent confidence interval:  
 0.8600472 1.5456671  
sample estimates:  
mean of x  
 1.202857
```

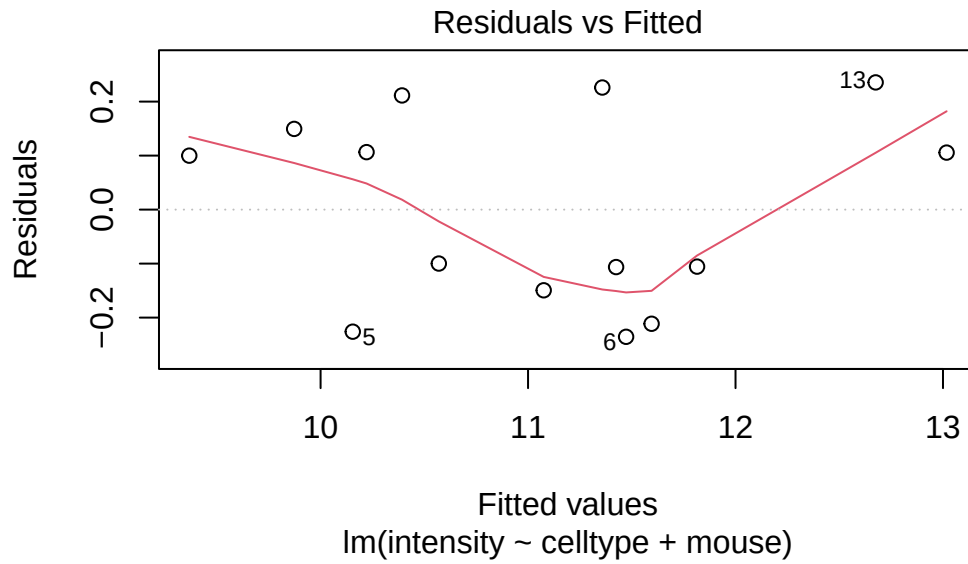
Randomized complete block analysis

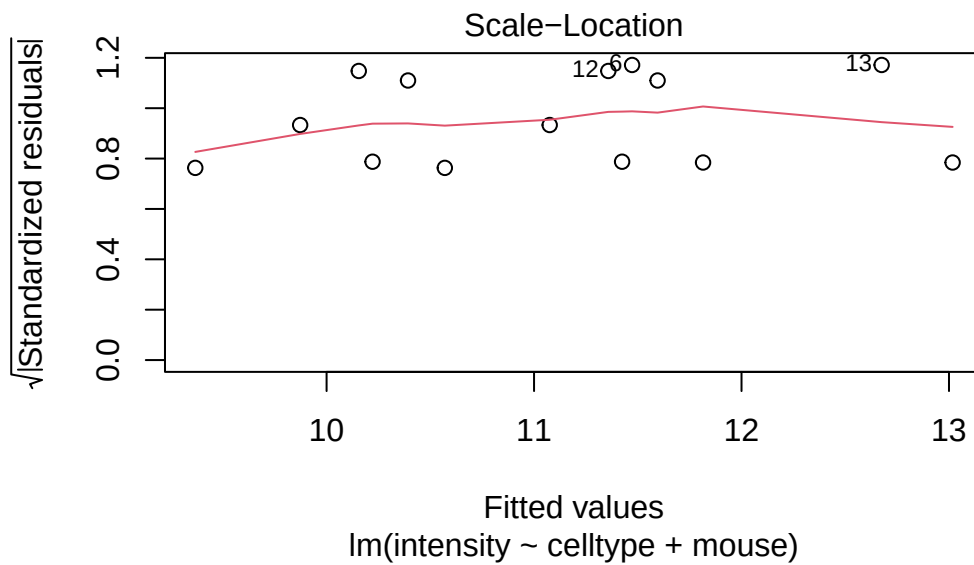
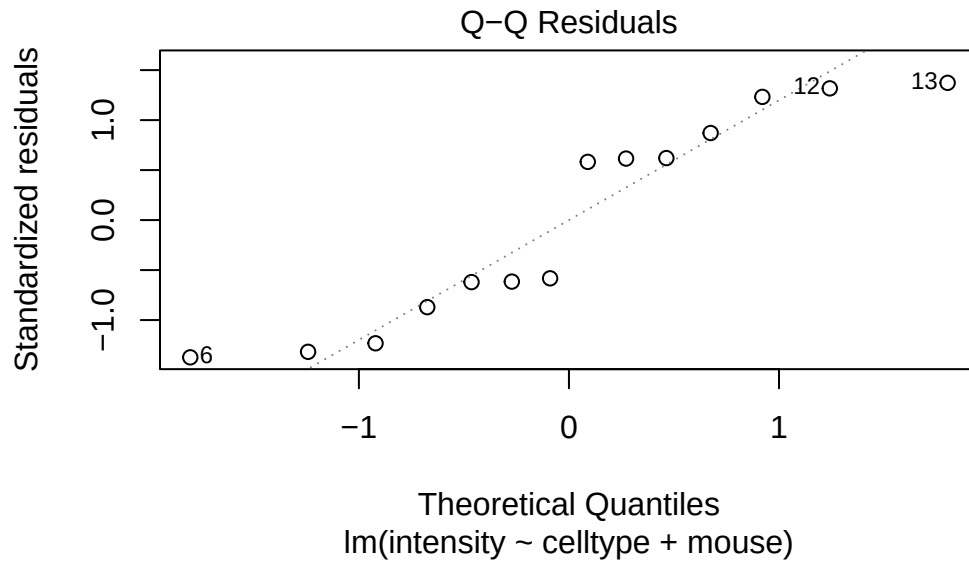
We can also analyse the design using a linear model, i.e. with

- a main effect for cell type and
- a main effect for the block factor mouse

Because we have measured the two cell types in every mouse, we can thus estimate the average log2-intensity of the protein in the T-cells for each mouse.

```
lmRCB <- lm(intensity ~ celltype + mouse, mouse)
plot(lmRCB, which = c(1, 2, 3))
```





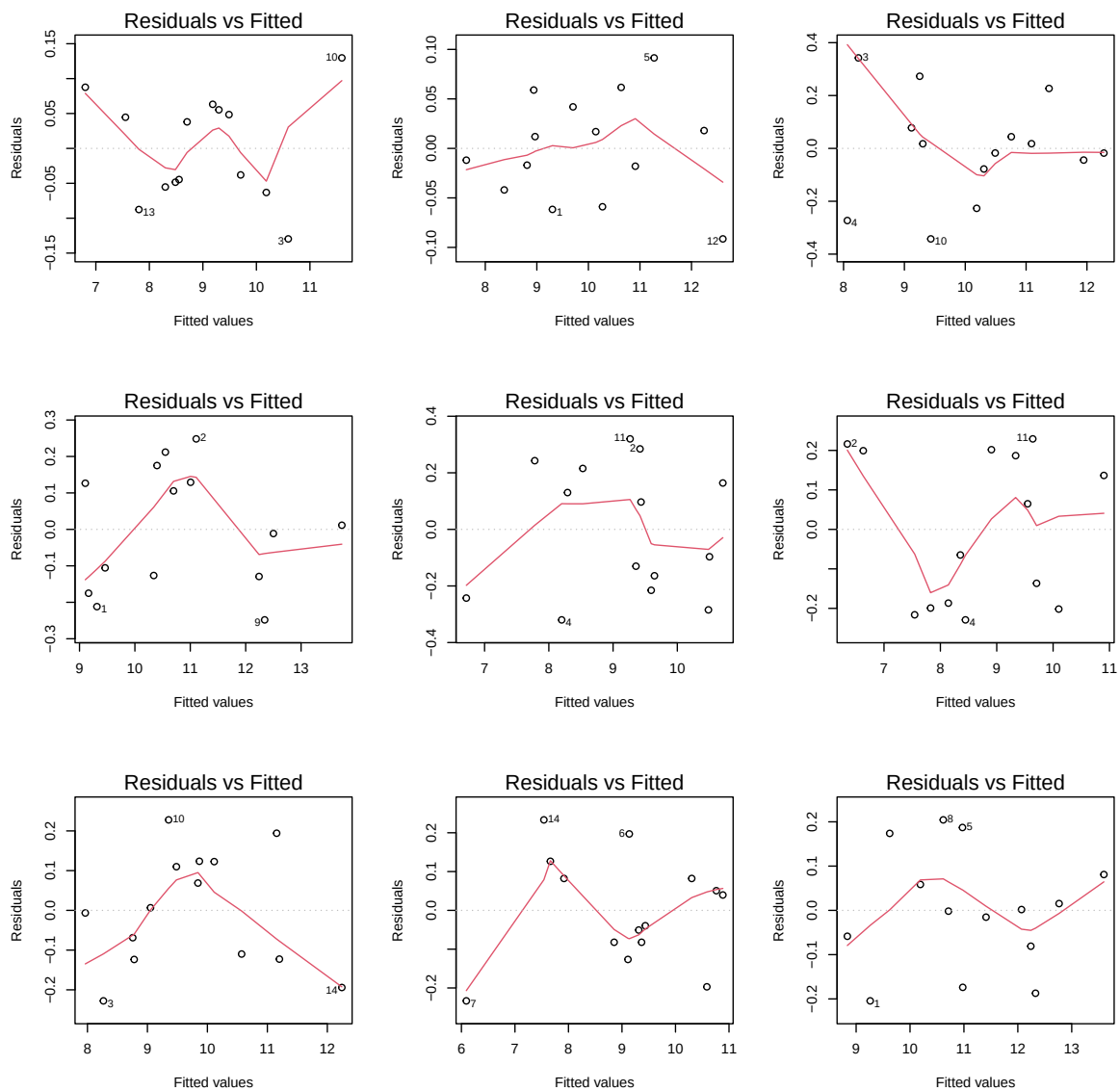
If you have doubts on that your data violates the assumptions you can always simulate data from a model with similar effects as yours but where are distributional assumptions hold and compare the residual plots.

```

design <- model.matrix(intensity ~ celltype + mouse, mouse)
sigmaMouse <- sqrt(car::Anova(lmRCB, type = "III")["mouse", "Sum Sq"] / car::Anova(lmRCB, ty
betas <- lmRCB$coefficients
nMouse <- mouse$mouse |>
  unique() |>
  length()

par(mfrow = c(3, 3))
for (i in 1:9)
{
  mouseEffect <- rnorm(nMouse, sd = sigmaMouse)
  betasMouse <- mouseEffect[-1] - mouseEffect[1]
  betas[-c(1:2)] <- betasMouse
  ysim <- design %*% betas + rnorm(nrow(design), sd = sigma(lmRCB))
  plot(lm(ysim ~ -1 + design), which = 1)
}

```



The deviations seen in our plot are in line with what those observed in simulations under the model assumptions. Hence, we can use the model for statistical inference.

```
anovaRCB <- car::Anova(lmRCB, type = "III")
summary(lmRCB)
```

Call:

```
lm(formula = intensity ~ celltype + mouse, data = mouse)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.2356	-0.1387	0.0000	0.1387	0.2356

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	9.3671	0.1981	47.277	6.00e-09	***
celltypeTreg	1.2029	0.1401	8.586	0.000137	***
mousem2	0.5055	0.2621	1.929	0.102036	
mousem3	1.0255	0.2621	3.913	0.007869	**
mousem4	0.8545	0.2621	3.260	0.017245	*
mousem5	0.7880	0.2621	3.006	0.023809	*
mousem6	2.1055	0.2621	8.033	0.000199	***
mousem7	2.4475	0.2621	9.338	8.55e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2621 on 6 degrees of freedom

Multiple R-squared: 0.9717, Adjusted R-squared: 0.9388

F-statistic: 29.47 on 7 and 6 DF, p-value: 0.000309

```
t.test(delta ~1 , mouseWide)
```

One Sample t-test

data: delta

t = 8.5858, df = 6, p-value = 0.0001372

alternative hypothesis: true mean is not equal to 0

95 percent confidence interval:

0.8600472 1.5456671

sample estimates:

mean of x

1.202857

```
anovaRCB
```

Anova Table (Type III tests)

Response: intensity

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	153.549	1	2235.154	6.002e-09	***
celltype	5.064	1	73.715	0.0001372	***
mouse	9.107	6	22.093	0.0007601	***
Residuals	0.412	6			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Notice that

1. the estimate, se, t-test statistic and p-value for the celltype effect of interest is the same as in the paired t-test!
2. the anova analysis shows that we model the total variability in the protein expression in T-cells using variability due to the cell type (CT), the variability from mouse to mouse (M) and residual variability (R) within mouse that we cannot explain. Indeed,

$$\begin{aligned} \text{SSTot} &= \text{SSCT} + \text{SSM} + \text{SSE} \\ 14.6 &= 5.1 + 9.1 + 0.4 \end{aligned}$$

So the celltype and the mouse effect explain respectively

$$\begin{aligned} \frac{\text{SSCT}}{\text{SSTot}} \times 100 & \quad \frac{\text{SSM}}{\text{SSTot}} \times 100 \\ 34.7 & \quad 62.4 \end{aligned}$$

percent of the variability in the protein expression values and

$$\frac{\text{SSE}}{\text{SSTot}} \times 100 = 2.8$$

percent cannot be explained.

Note, that

- the variability from mouse to mouse is the largest source of variability in the model,
- This variability can be estimated with the RCB design and
- can thus be isolated from the residual variability
- leading to a much higher precision on the estimate of the average log2 FC between regulatory and conventional T-cells that what would be obtained with a CRD!

Note, that the RCB also has to sacrifice a number of degrees of freedom to estimate the mouse effect, here 6 DF.

Hence, the power gain of the RCB is a trade-off between the variability that can be explained by the block effect and the loss in DF.

→ We were able to isolate the variance in expression between animals/blocks from our analysis!

→ This reduces the variance of the residuals and leads to a power gain if the variability between mice/blocks is large.

Also note that,

$$\hat{\sigma}^2 = \frac{SSE}{n - p} = \frac{SSTot - SSM - SSCT}{n - p}$$

- Hence, blocking is beneficial if the reduction in sum of squares of the residuals is large compared to the degrees of freedom that are sacrificed.
- Thus if SSM can explain a large part of the total variability.

Further, the degrees of freedom affect the t-distribution that is used for inference, which has broader tails if the residual degrees of freedom $n - p$ are getting smaller.

Power gain of an RCB vs CRD

In this section we will subset the original data in two experiments:

- A randomized complete block design with three mice
- A completely randomized design with six mice but where we only measure one cell type in each mouse.

```
set.seed(859)
mRcb <- mouse |>
  pull(mouse) |>
  unique() |>
  sample(size = 3)

rcbSmall <- mouse |> filter(mouse %in% mRcb)
rcbSmall
```

```
# A tibble: 6 x 3
  celltype mouse intensity
  <chr>    <chr>    <dbl>
1 Tcon     m2         10.0
2 Tcon     m3         10.6
3 Tcon     m7         11.7
4 Treg     m2         10.9
5 Treg     m3         11.4
6 Treg     m7         13.1
```

```
mCrd <- mouse |>
  pull(mouse) |>
  unique() |>
  sample(size = 6)

crdSmall <-
  bind_rows(
    mouse |>
      filter(mouse %in% mCrd[1:3]) |>
      filter(celltype == "Tcon"),
    mouse |>
      filter(mouse %in% mCrd[-(1:3)]) |>
      filter(celltype == "Treg")
  )
crdSmall
```

```
# A tibble: 6 x 3
  celltype mouse intensity
  <chr>    <chr>    <dbl>
1 Tcon     m2         10.0
2 Tcon     m4         10.3
3 Tcon     m6         11.2
4 Treg     m1         10.5
5 Treg     m3         11.4
6 Treg     m7         13.1
```

So in both experiments we need to do six mass spectrometry runs.

```
lmCRDSmall <- lm(intensity ~ celltype, crdSmall)
summary(lmCRDSmall)
```

Call:

```
lm(formula = intensity ~ celltype, data = crdSmall)
```

Residuals:

1	2	3	4	5	6
-0.507	-0.201	0.708	-1.189	-0.275	1.464

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.5290	0.6077	17.326	6.51e-05 ***
celltypeTreg	1.1300	0.8594	1.315	0.259

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.053 on 4 degrees of freedom

Multiple R-squared: 0.3018, Adjusted R-squared: 0.1272

F-statistic: 1.729 on 1 and 4 DF, p-value: 0.2589

```
anova(lmCRDSmall)
```

Analysis of Variance Table

Response: intensity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
celltype	1	1.9153	1.9154	1.7289	0.2589
Residuals	4	4.4314	1.1078		

```
lmRCBSmall <- lm(intensity ~ celltype + mouse, rcbSmall)
anova(lmRCBSmall)
```

Analysis of Variance Table

Response: intensity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
celltype	1	1.5996	1.59960	28.335	0.03353 *
mouse	2	4.0426	2.02128	35.805	0.02717 *
Residuals	2	0.1129	0.05645		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
summary(lmRCBsmall)
```

Call:

```
lm(formula = intensity ~ celltype + mouse, data = rcbSmall)
```

Residuals:

1	2	3	4	5	6
0.06433	0.12633	-0.19067	-0.06433	-0.12633	0.19067

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	9.9577	0.1940	51.329	0.000379	***
celltypeTreg	1.0327	0.1940	5.323	0.033527	*
mousem3	0.5200	0.2376	2.189	0.160094	
mousem7	1.9420	0.2376	8.173	0.014641	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2376 on 2 degrees of freedom

Multiple R-squared: 0.9804, Adjusted R-squared: 0.951

F-statistic: 33.32 on 3 and 2 DF, p-value: 0.02928

Note, that

- we are able to pick up the upregulation between regulatory T-cells and ordinary T-cells with the RCB but not with the CRD.
- the standard error of the $\log_2 FC_{Treg-Tcon}$ estimate is a factor 4.4 smaller for the RCB design!

A poor data analysis who forgets about the blocking will be back at square one:

```
wrongRbc <- lm(intensity ~ celltype, rcbSmall)
anova(wrongRbc)
```

Analysis of Variance Table

Response: intensity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
celltype	1	1.5996	1.5996	1.5398	0.2825
Residuals	4	4.1555	1.0389		

```
summary(wrongRbc)
```

Call:

```
lm(formula = intensity ~ celltype, data = rcbSmall)
```

Residuals:

```
      1      2      3      4      5      6  
-0.7563 -0.1743  0.9307 -0.8850 -0.4270  1.3120
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)  
(Intercept)   10.7783     0.5885   18.316 5.23e-05 ***  
celltypeTreg    1.0327     0.8322    1.241   0.282  
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.019 on 4 degrees of freedom

Multiple R-squared: 0.2779, Adjusted R-squared: 0.09743

F-statistic: 1.54 on 1 and 4 DF, p-value: 0.2825

So, the block to block variability is then absorbed in the variance estimator of the residual.

Of course, we are never allowed to analyse an RCB with a model for a CRD (without block factor) because blocking imposes a randomization restriction: we randomize the treatment within block.

Power gain of blocking?

Power for completely randomized design

```
varBetweenPlusWithin <- sum(car::Anova(lmRCB, type = "III")[c("mouse", "Residuals"), "Sum Sq"]  
  
alpha <- 0.05  
nSim <- 20000  
b0 <- 0  
sd <- sqrt(varBetweenPlusWithin)  
ns <- c(3, 7)  
deltas <- lmRCB$coefficients["celltypeTreg"]
```

```
L <- limma::makeContrasts("celltypeTreg", levels = c("(Intercept)", "celltypeTreg"))
```

Renaming (Intercept) to Intercept

```
powerFast <- matrix(NA, nrow = length(ns) * length(deltas), ncol = 3) |> as.data.frame()
names(powerFast) <- c("b1", "n", "power")

i <- 0
for (n in ns)
{
  n1 <- n2 <- n

  ### Simulation
  predictorData <- data.frame(celltype = rep(c("Tcon", "Treg"), c(n1, n2)) |> as.factor())
  design <- model.matrix(~celltype, predictorData)

  for (b1 in deltas)
  {
    ySim <- rnorm(nrow(predictorData) * nSim, sd = sd)
    dim(ySim) <- c(nrow(predictorData), nSim)
    ySim <- ySim + c(design %*% c(b0, b1))
    ySim <- t(ySim)

    ### Fitting
    fitAll <- limma::lmFit(ySim, design)

    ### Inference
    varUnscaled <- c(t(L) %*% fitAll$cov.coefficients %*% L)
    contrasts <- fitAll$coefficients %*% L
    seContrasts <- varUnscaled^.5 * fitAll$sigma
    tstats <- contrasts / seContrasts
    pvals <- pt(abs(tstats), fitAll$df.residual, lower.tail = FALSE) * 2

    i <- i + 1
    powerFast[i, ] <- c(b1, n, mean(pvals < alpha))
  }
}
powerFast
```

```
b1 n power
```

```
1 1.202857 3 0.24675
2 1.202857 7 0.64355
```

Because we have a simple 2 group comparison we can also calculate the power using the `power.t.test` function.

```
power.t.test(n = 3, delta = lmRCB$coefficients["celltypeTreg"], sd = sqrt(varBetweenPlusWithin))
```

Two-sample t test power calculation

```
      n = 3
    delta = 1.202857
      sd = 0.8906339
sig.level = 0.05
  power = 0.2477638
alternative = two.sided
```

NOTE: n is number in *each* group

```
power.t.test(n = 7, delta = lmRCB$coefficients["celltypeTreg"], sd = sqrt(varBetweenPlusWithin))
```

Two-sample t test power calculation

```
      n = 7
    delta = 1.202857
      sd = 0.8906339
sig.level = 0.05
  power = 0.6411438
alternative = two.sided
```

NOTE: n is number in *each* group

Power for randomized complete block design

```
alpha <- 0.05
nSim <- 20000
b0 <- 0
```

```

sd <- sigma(lmRCB)
sdMouse <- sqrt(car::Anova(lmRCB)["mouse", "Sum Sq"] / car::Anova(lmRCB)["mouse", "Df"])
ns <- c(3, 7)
deltas <- lmRCB$coefficients["celltypeTreg"]

powerFastBlocking <- matrix(NA, nrow = length(ns) * length(deltas), ncol = 3) |> as.data.frame()
names(powerFastBlocking) <- c("b1", "n", "power")

i <- 0
for (n in ns)
{
  ### Simulation
  predictorData <- data.frame(celltype = rep(c("Tcon", "Treg"), each = n) |> as.factor(), mouse = rep(0, n))
  design <- model.matrix(~ celltype + mouse, predictorData)
  L <- limma::makeContrasts("celltypeTreg", levels = colnames(design))

  for (b1 in deltas)
  {
    ySim <- rnorm(nrow(predictorData) * nSim, sd = sd)
    dim(ySim) <- c(nrow(predictorData), nSim)
    mouseEffect <- rnorm(n, sd = sdMouse)
    betasMouse <- mouseEffect[-1] - mouseEffect[1]
    ySim <- ySim + c(design %*% c(b0, b1, betasMouse))
    ySim <- t(ySim)

    ### Fitting
    fitAll <- limma::lmFit(ySim, design)

    ### Inference
    varUnscaled <- c(t(L) %*% fitAll$cov.coefficients %*% L)
    contrasts <- fitAll$coefficients %*% L
    seContrasts <- varUnscaled^.5 * fitAll$sigma
    tstats <- contrasts / seContrasts
    pvals <- pt(abs(tstats), fitAll$df.residual, lower.tail = FALSE) * 2

    i <- i + 1
    powerFastBlocking[i, ] <- c(b1, n, mean(pvals < alpha))
  }
}

```

Renaming (Intercept) to Intercept
Renaming (Intercept) to Intercept

```
powerFastBlocking
```

```
      b1 n    power
1 1.202857 3 0.79505
2 1.202857 7 1.00000
```

Note, that the power is indeed much larger for the randomized complete block design. Both for the designs with 6 and 14 mass spectrometer runs.

Because we have an RCB with a block size of 2 (paired design) we can also calculate the power using the `power.t.test` function with `type = "one.sample"` and `sd` equal to the standard deviation of the difference.

```
power.t.test(n = 3, delta = mean(mouseWide$delta), sd = sd(mouseWide$delta))
```

Two-sample t test power calculation

```
      n = 3
delta = 1.202857
sd = 0.3706672
sig.level = 0.05
power = 0.8389961
alternative = two.sided
```

NOTE: n is number in *each* group

```
power.t.test(n = 7, delta = mean(mouseWide$delta), sd = sd(mouseWide$delta))
```

Two-sample t test power calculation

```
      n = 7
delta = 1.202857
sd = 0.3706672
sig.level = 0.05
power = 0.999826
```

```
alternative = two.sided
```

NOTE: n is number in *each* group

Note, that the power is slightly different because for the power.t.test function we conditioned on the mice from our study. While in the simulation study we generated data for new mice by simulating the mouse effect from a normal distribution.

Impact of the amount of variability that the blocking factor explains on the power?

We will vary the block effect explains

$$\frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2} = 1 - \frac{\sigma_{\text{within}}^2}{\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2}$$

So in our example that is the ratio between the variability between the mice and the sum of the variability between and within the mice. Note, that the within mouse variability was the variance of the errors of the RCB. The ratio for our experiment equals

```
varBetweenPlusWithin <- sum(car::Anova(lmRCB, type = "III")[c("mouse", "Residuals"), "Sum Sq"]  
varWithin <- car::Anova(lmRCB)[c("Residuals", "Sum Sq"), "Sum Sq"] / car::Anova(lmRCB)[c("Residuals", "Df"), "Df"]  
varBetweenPlusWithin
```

```
[1] 0.7932287
```

```
varWithin
```

```
[1] 0.06869707
```

```
1 - varWithin / varBetweenPlusWithin
```

```
[1] 0.9133956
```

```
alpha <- 0.05  
nSim <- 20000  
b0 <- 0  
varBetweenPlusWithin <- sum(car::Anova(lmRCB, type = "III")[c("mouse", "Residuals"), "Sum Sq"]  
  
ns <- c(3, 7)
```

```

deltas <- lmRCB$coefficients["celltypeTreg"]

fracVars <- seq(0, .95, .05)

powerFastBlockingLow <- matrix(NA, nrow = length(ns) * length(fracVars), ncol = 3) |> as.data.frame()
names(powerFastBlockingLow) <- c("fracVars", "n", "power")

i <- 0

for (n in ns)
{
  ### Simulation
  predictorData <- data.frame(celltype = rep(c("Tcon", "Treg"), each = n) |> as.factor(), mouse = rep(1:n, each = 2))
  design <- model.matrix(~ celltype + mouse, predictorData)
  L <- limma::makeContrasts("celltypeTreg", levels = colnames(design))
  for (fracVar in fracVars)
  {
    sd <- sqrt(varBetweenPlusWithin * (1 - fracVar))
    sdMouse <- sqrt(varBetweenPlusWithin * fracVar)
    for (b1 in deltas)
    {
      ySim <- rnorm(nrow(predictorData) * nSim, sd = sd)
      dim(ySim) <- c(nrow(predictorData), nSim)
      mouseEffect <- rnorm(n, sd = sdMouse)
      betasMouse <- mouseEffect[-1] - mouseEffect[1]
      ySim <- ySim + c(design %*% c(b0, b1, betasMouse))
      ySim <- t(ySim)

      ### Fitting
      fitAll <- limma::lmFit(ySim, design)

      ### Inference
      varUnscaled <- c(t(L) %*% fitAll$cov.coefficients %*% L)
      contrasts <- fitAll$coefficients %*% L
      seContrasts <- varUnscaled^.5 * fitAll$sigma
      tstats <- contrasts / seContrasts
      pvals <- pt(abs(tstats), fitAll$df.residual, lower.tail = FALSE) * 2

      i <- i + 1
      powerFastBlockingLow[i, ] <- c(fracVar, n, mean(pvals < alpha))
    }
  }
}

```

```

    }
  }
}

```

Renaming (Intercept) to Intercept
 Renaming (Intercept) to Intercept

```
powerFastBlockingLow
```

	fracVars	n	power
1	0.00	3	0.16935
2	0.05	3	0.17575
3	0.10	3	0.18705
4	0.15	3	0.19050
5	0.20	3	0.19445
6	0.25	3	0.20330
7	0.30	3	0.20745
8	0.35	3	0.22805
9	0.40	3	0.24110
10	0.45	3	0.26075
11	0.50	3	0.27305
12	0.55	3	0.29500
13	0.60	3	0.31615
14	0.65	3	0.34370
15	0.70	3	0.38995
16	0.75	3	0.44120
17	0.80	3	0.51210
18	0.85	3	0.60890
19	0.90	3	0.75210
20	0.95	3	0.93550
21	0.00	7	0.55920
22	0.05	7	0.58150
23	0.10	7	0.60545
24	0.15	7	0.62625
25	0.20	7	0.65755
26	0.25	7	0.68370
27	0.30	7	0.71110
28	0.35	7	0.73940
29	0.40	7	0.77070
30	0.45	7	0.80430
31	0.50	7	0.85040

```

32     0.55 7 0.88100
33     0.60 7 0.90875
34     0.65 7 0.94025
35     0.70 7 0.96775
36     0.75 7 0.98600
37     0.80 7 0.99525
38     0.85 7 0.99955
39     0.90 7 1.00000
40     0.95 7 1.00000

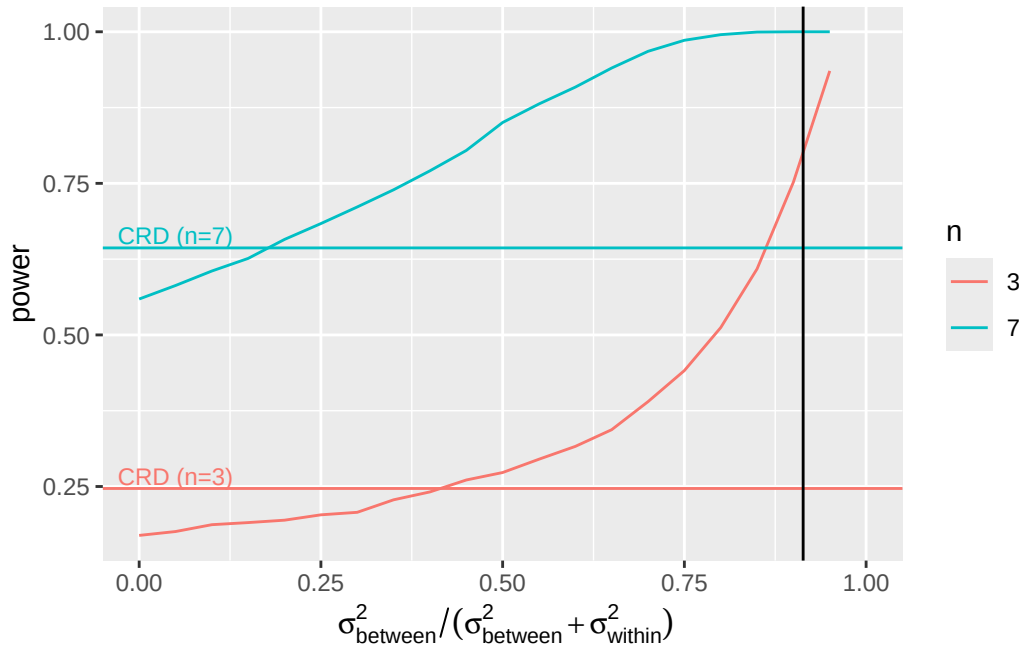
```

```

gg_color_hue <- function(n) {
  hues <- seq(15, 375, length = n + 1)
  hcl(h = hues, l = 65, c = 100)[1:n]
}
cols <- gg_color_hue(2)

powerFastBlockingLow |>
  as.data.frame() |>
  mutate(n = as.factor(n)) |>
  ggplot(aes(fracVars, power, group = n, color = n)) +
  geom_line() +
  geom_hline(yintercept = powerFast |> filter(n == 3) |> pull(power), color = cols[1]) +
  annotate("text",
    label = "CRD (n=3)",
    x = 0.05, y = powerFast |> filter(n == 3) |> pull(power) + .02, size = 3, colour = cols[1]) +
  geom_hline(yintercept = powerFast |> filter(n == 7) |> pull(power), color = cols[2]) +
  annotate("text",
    label = "CRD (n=7)",
    x = 0.05, y = powerFast |> filter(n == 7) |> pull(power) + .02, size = 3, colour = cols[2]) +
  xlab(expression(~ sigma[between]^2 / (sigma[between]^2 + sigma[within]^2))) +
  geom_vline(xintercept = 1 - varWithin / varBetweenPlusWithin) +
  xlim(0, 1)

```



- So if the variance that is explained by the block effect is small you will lose power as compared to the analysis with a CRD design. Indeed, then
 - SSE does not reduce much and
 - $n_{\text{blocks}} - 1$ degrees of freedom have been sacrificed.
- As soon as the block effect explains a large part of the variability it is very beneficial to use a randomized complete block design!
- Note, that the same number of mass spectrometry runs have to be done for both the RCB and CRD design. However, for the RCB we only need half of the mice.

Penicillin example

The production of penicillin corn steep liquor is used. Corn steep liquor is produced in blends and there is a considerable variability between the blends. Suppose that

- four competing methods have to be evaluated to produce penicillin (A-D),
- one blend is sufficient for four runs of a penicillin batch reactor and
- the 20 runs can be scheduled for the experiment.

How would you assign the methods to the blends.

```
data(penicillin, package = "faraway")
table(penicillin$blend, penicillin$treat)
```

```
      A B C D
Blend1 1 1 1 1
Blend2 1 1 1 1
Blend3 1 1 1 1
Blend4 1 1 1 1
Blend5 1 1 1 1
```

Data

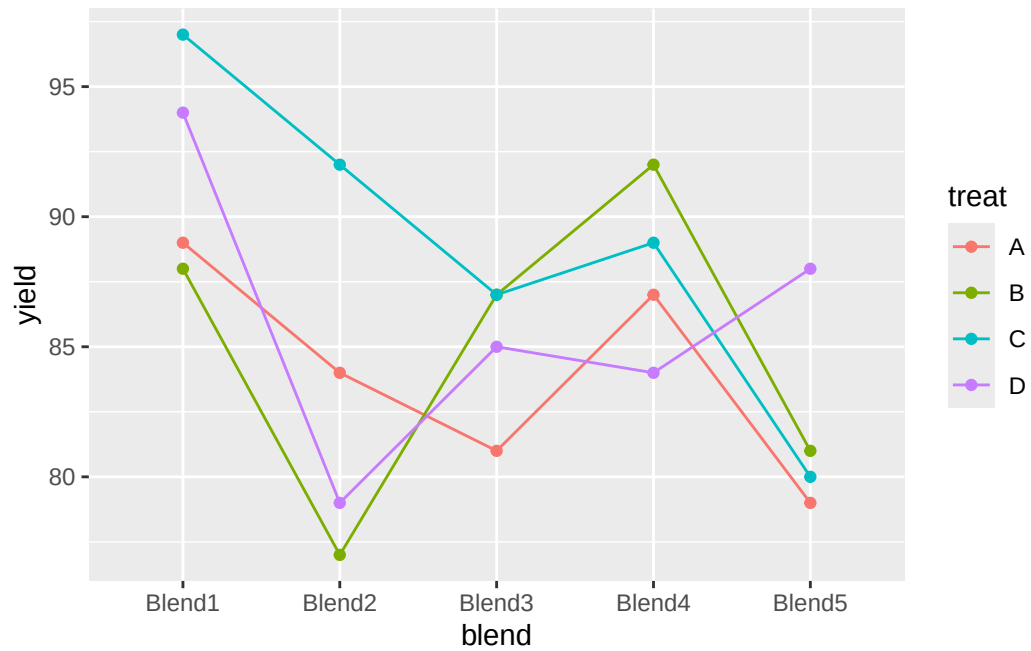
```
head(penicillin)
```

```
  treat blend yield
1     A Blend1    89
2     B Blend1    88
3     C Blend1    97
4     D Blend1    94
5     A Blend2    84
6     B Blend2    77
```

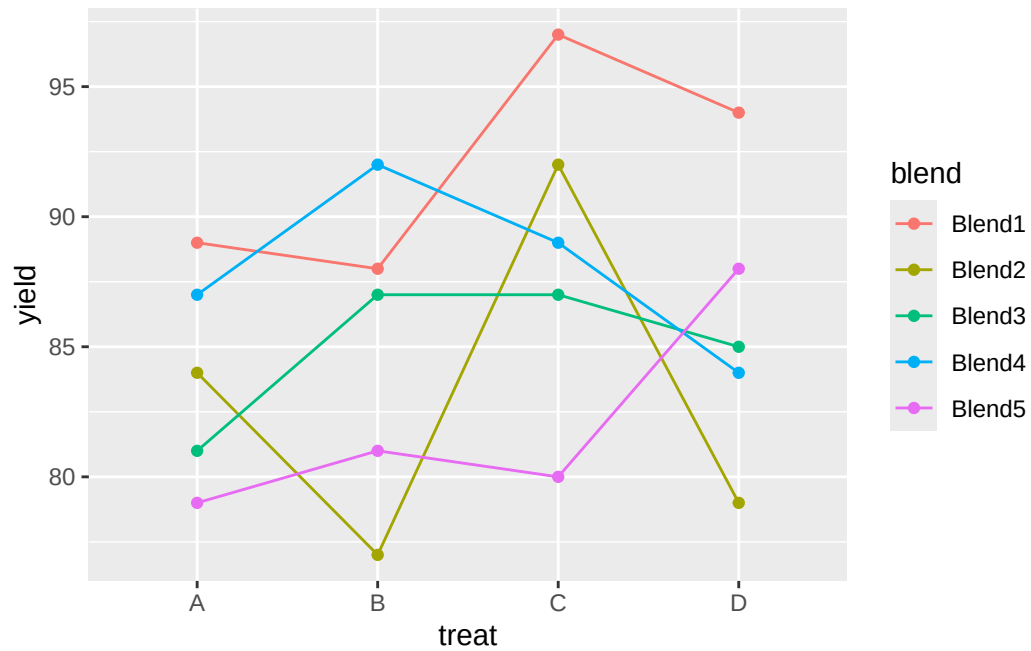
```
matrix(penicillin$yield, nrow = 5, ncol = 4, byrow = TRUE, dimnames = list(levels(penicillin$blend), levels(penicillin$treat)))
```

```
      A B C D
Blend1 89 88 97 94
Blend2 84 77 92 79
Blend3 81 87 87 85
Blend4 87 92 89 84
Blend5 79 81 80 88
```

```
penicillin |>
  ggplot(aes(x = blend, y = yield, group = treat, color = treat)) +
  geom_line() +
  geom_point()
```



```
penicillin |>
  ggplot(aes(x = treat, y = yield, group = blend, color = blend)) +
  geom_line() +
  geom_point()
```

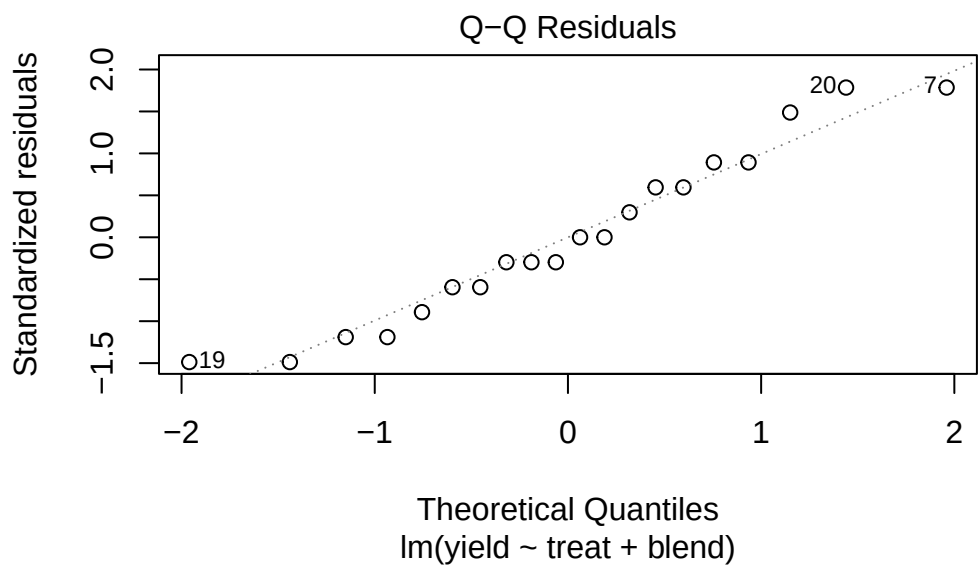
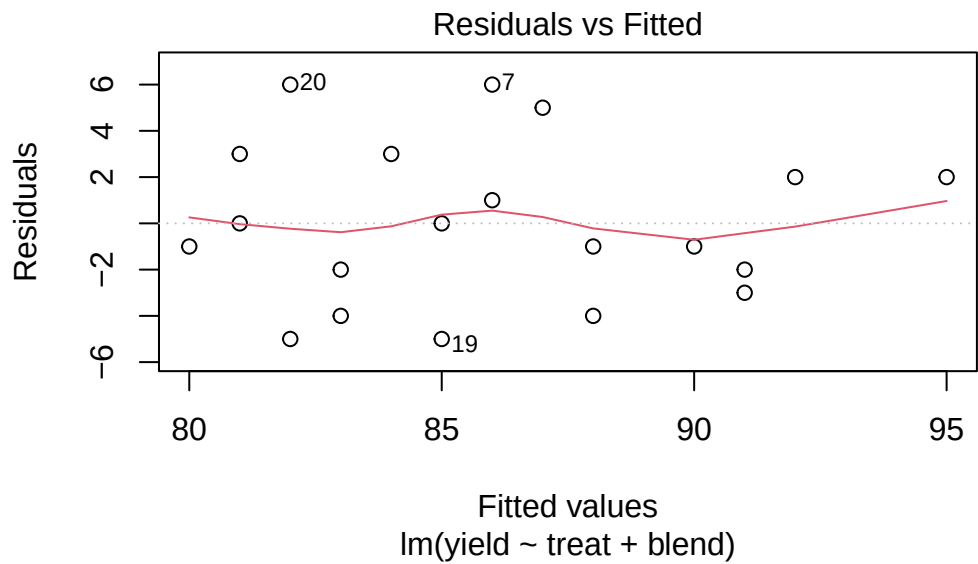


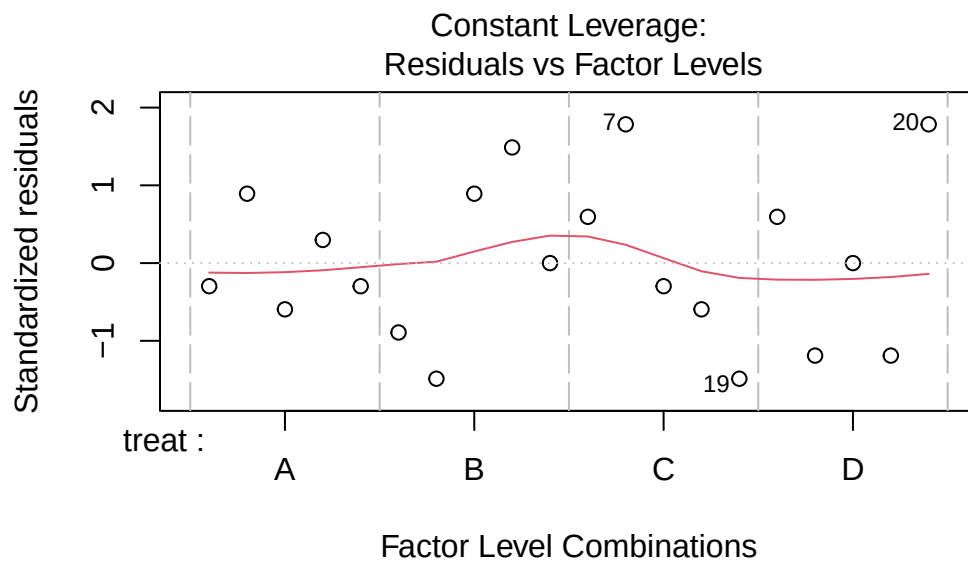
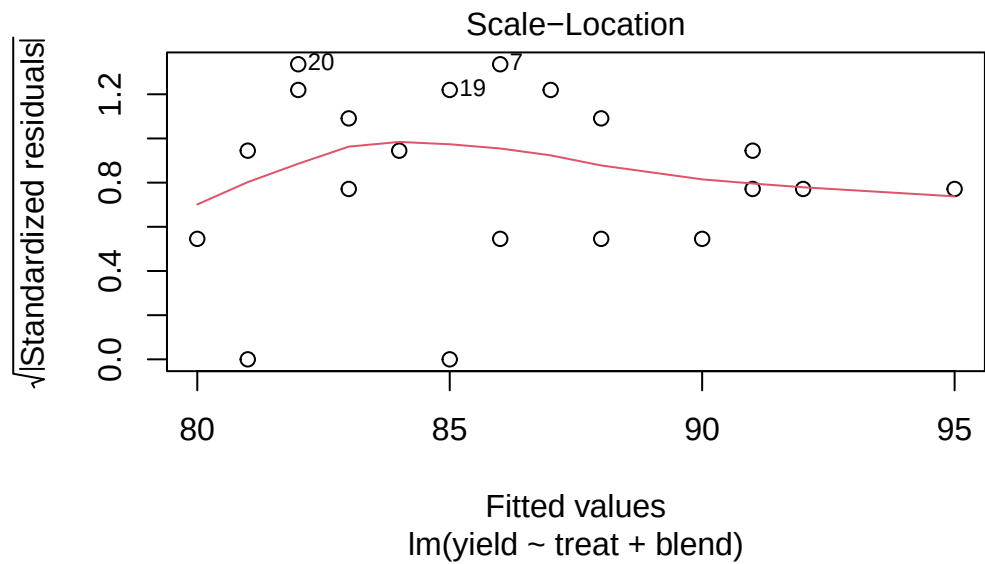
Analysis

We analyse the yield using

- a factor for blend and
- a factor for treatment.

```
lmPen <- lm(yield ~ treat + blend, data = penicillin)
plot(lmPen)
```





```
car::Anova(lmPen, type = "III")
```

Anova Table (Type III tests)

```

Response: yield
      Sum Sq Df    F value    Pr(>F)
(Intercept) 20250 1 1075.2212 4.098e-13 ***
treat         70 3    1.2389  0.33866
blend        264 4    3.5044  0.04075 *
Residuals    226 12
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

We conclude that the effect of the treatment on the penicillin yield is not significant at the 5% level of significance ($p = 0.34$).

We also observe that there is a large effect of the blend on the yield. Blend explains about 47.1% of the variability in the penicillin yield.

Pseudo-replication

A study on the facultative pathogen *Francisella tularensis* was conceived by Ramond et al. (2015) [12].

- *F. tularensis* enters the cells of its host by phagocytosis.
- The authors showed that *F. tularensis* is arginine deficient and imports arginine from the host cell via an arginine transporter, ArgP, in order to efficiently escape from the phagosome and reach the cytosolic compartment, where it can actively multiply.
- In their study, they compared the proteome of wild type *F. tularensis* (WT) to ArgP-gene deleted *F. tularensis* (knock-out, D8).
- The sample for each bio-rep was run in technical triplicate on the mass-spectrometer.
- We will use data for the protein 50S ribosomal protein L5 [A0Q4J5](#)

Data exploration

```
franc <- read_tsv("https://raw.githubusercontent.com/statOmics/PSLSDData/main/francisellaA0Q4J5")
```

```

Rows: 18 Columns: 3
-- Column specification -----
Delimiter: "\t"
chr (2): genotype, biorep
dbl (1): intensityLog2

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```
franc
```

```

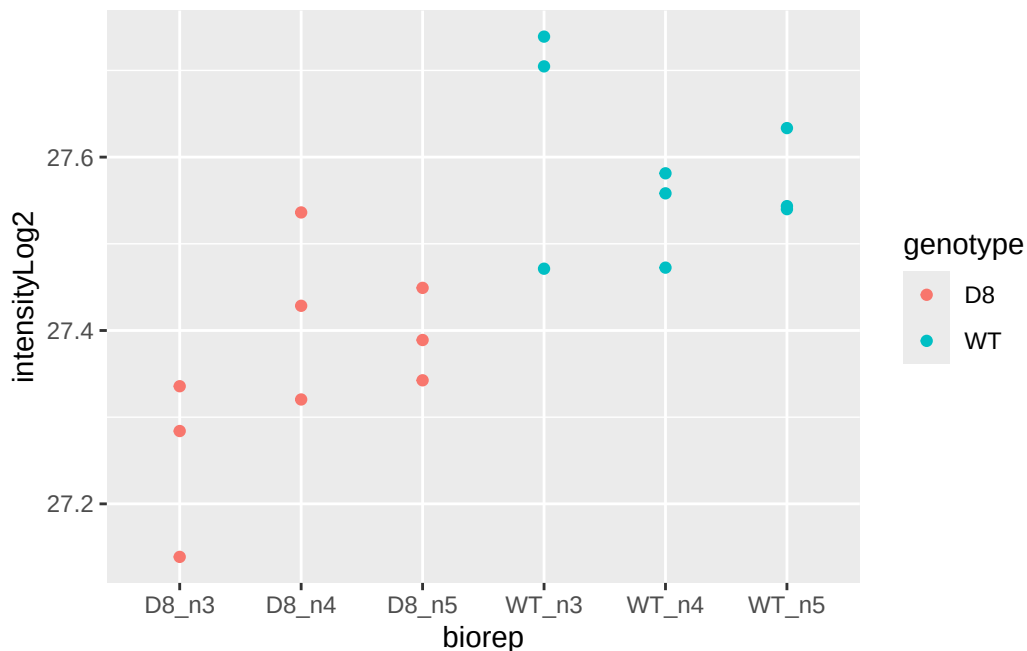
# A tibble: 18 x 3
  genotype biorep intensityLog2
  <chr>     <chr>         <dbl>
1 WT       WT_n3          27.7
2 WT       WT_n3          27.5
3 WT       WT_n3          27.7
4 WT       WT_n4          27.5
5 WT       WT_n4          27.6
6 WT       WT_n4          27.6
7 WT       WT_n5          27.5
8 WT       WT_n5          27.5
9 WT       WT_n5          27.6
10 D8      D8_n3          27.1
11 D8      D8_n3          27.3
12 D8      D8_n3          27.3
13 D8      D8_n4          27.3
14 D8      D8_n4          27.4
15 D8      D8_n4          27.5
16 D8      D8_n5          27.3
17 D8      D8_n5          27.4
18 D8      D8_n5          27.4

```

```

franc |>
  ggplot(aes(biorep, intensityLog2, color = genotype)) +
  geom_point()

```



- Response?
- Experimental unit?
- Observational unit?
- Factors?

→ Pseudo-replication, randomisation to bio-repeat and each bio-repeat measured in technical triplicate. → If we would analyse the data using a linear model based on each measured intensity, we would act as if we had sampled 18 bio-repeats. → Effect of interest has to be assessed between bio-repeats. So block analysis not possible!

If the same number of pseudo-replicates/technical replicates are available for each bio-repeat:

- average first over bio-repeats to obtain independent measurements
- averages will then have the same precision
- assess effect of treatment using averages
- **Caution: never summarize over bio-repeats/experimental units**

```
lmBiorep <- lm(intensityLog2 ~ -1 + biorep, franc)
lmBiorep
```

Call:

```
lm(formula = intensityLog2 ~ -1 + biorep, data = franc)
```

Coefficients:

biorepD8_n3	biorepD8_n4	biorepD8_n5	biorepWT_n3	biorepWT_n4	biorepWT_n5
27.25	27.43	27.39	27.64	27.54	27.57

```
francSum <- data.frame(genotype = rep(c("D8", "WT"), each = 3) |> as.factor()) |> relevel("WT", first = TRUE)
francSum
```

	genotype	intensityLog2
biorepD8_n3	D8	27.25286
biorepD8_n4	D8	27.42836
biorepD8_n5	D8	27.39357
biorepWT_n3	WT	27.63831
biorepWT_n4	WT	27.53734
biorepWT_n5	WT	27.57232

```
lmSum <- lm(intensityLog2 ~ genotype, francSum)
summary(lmSum)
```

Call:

```
lm(formula = intensityLog2 ~ genotype, data = francSum)
```

Residuals:

biorepD8_n3	biorepD8_n4	biorepD8_n5	biorepWT_n3	biorepWT_n4	biorepWT_n5
-0.10541	0.07010	0.03531	0.05566	-0.04531	-0.01034

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.58266	0.04333	636.581	3.65e-11 ***
genotypeD8	-0.22439	0.06128	-3.662	0.0215 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07505 on 4 degrees of freedom

Multiple R-squared: 0.7702, Adjusted R-squared: 0.7128

F-statistic: 13.41 on 1 and 4 DF, p-value: 0.02154

Wrong analysis

```
lmWrong <- lm(intensityLog2 ~ genotype, franc)
summary(lmWrong)
```

Call:

```
lm(formula = intensityLog2 ~ genotype, data = franc)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.21943	-0.04181	-0.01914	0.06537	0.17792

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	27.35826	0.03457	791.48	< 2e-16 ***
genotypeWT	0.22439	0.04888	4.59	0.000302 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1037 on 16 degrees of freedom

Multiple R-squared: 0.5684, Adjusted R-squared: 0.5414

F-statistic: 21.07 on 1 and 16 DF, p-value: 0.0003017

Note, that the analysis where we ignore that we have multiple technical repeats for each bio-repeat returns results that are much more significant because we act as if we have much more independent observations.

No Type I error control!

```
sigmaWithin <- sigma(lmBiorep)
sigmaBetween <- sigma(lmSum)
xBiorep <- model.matrix(~ -1 + biorep, franc)
xWrong <- model.matrix(~genotype, franc)

set.seed(2523)
nSim <- 1000
resWrong <- matrix(NA, nSim, 4) |> as.data.frame()
names(resWrong) <- c("Estimate", "Std. Error", "t value", "pvalue")
resCorrect <- resWrong
```

```

genotype <- franc$genotype
genotypeSum <- francSum$genotype
biorep <- franc$biorep

for (i in 1:nSim)
{
  biorepSim <- rnorm(ncol(xBiorep), sd = sigmaBetween)
  ySim <- xBiorep %*% biorepSim + rnorm(nrow(xBiorep), sd = sigmaWithin)
  ySum <- lm(ySim ~ biorep)$coefficient
  resWrong[i, ] <- summary(lm(ySim ~ genotype))$coefficient[2, ]
  resCorrect[i, ] <- summary(lm(ySum ~ genotypeSum))$coefficient[2, ]
}
mean(resCorrect$pvalue < 0.05)

```

```
[1] 0.042
```

```
mean(resWrong$pvalue < 0.05)
```

```
[1] 0.143
```

```

qplot(resCorrect$pvalue, geom = "histogram", boundary = c(0, 1)) +
  stat_bin(breaks = seq(0, 1, .1)) +
  xlab("pvalue") +
  ggtitle("Correct analysis")

```

Warning: `qplot()` was deprecated in ggplot2 3.4.0.

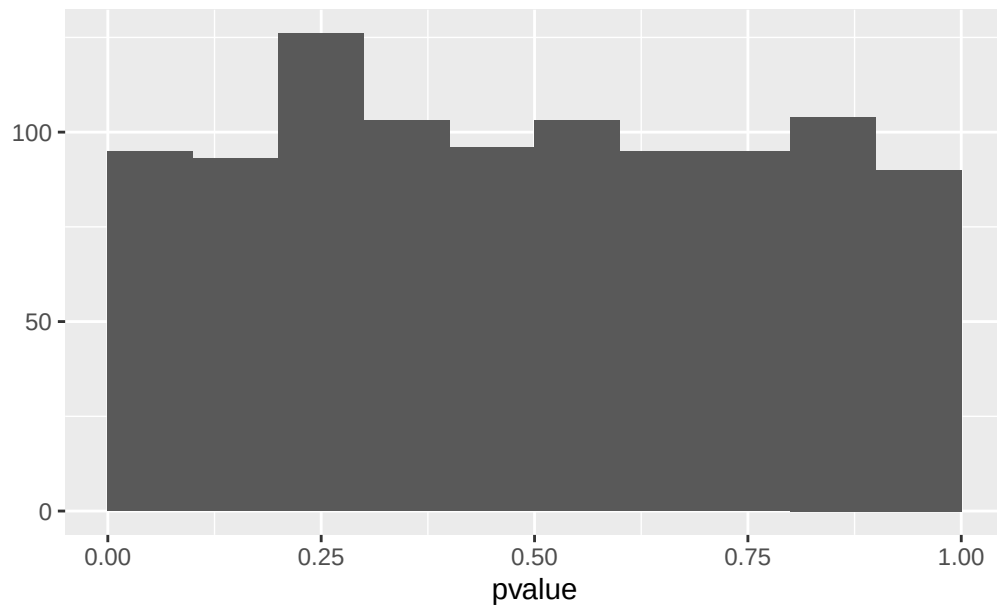
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Computation failed in `stat\_bin()`.

Caused by error in `compute\_bins()`:

! `boundary` must be a number or `NULL`, not a double vector.

Correct analysis



```
qplot(resWrong$pvalue, geom = "histogram", boundary = c(0, 1)) +  
  stat_bin(breaks = seq(0, 1, .1)) +  
  xlab("pvalue") +  
  ggtitle("Wrong analysis")
```

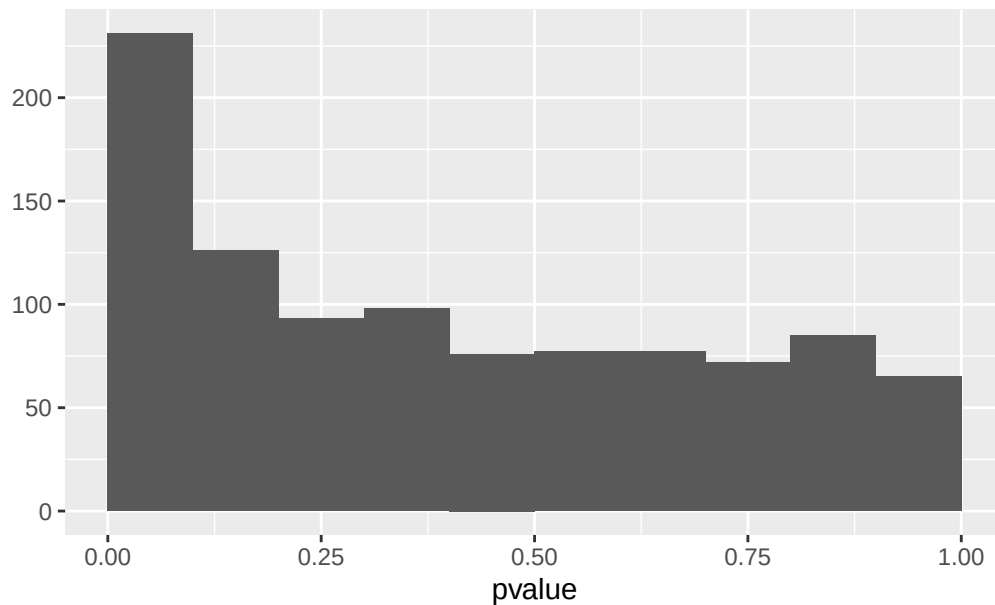
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

Warning: Computation failed in ``stat_bin()``.

Caused by error in ``compute_bins()``:

! ``boundary`` must be a number or ``NULL``, not a double vector.

Wrong analysis



- So we observe that the analysis that does not account for pseudo-replication is too liberal!
- The analysis that first summarizes over the technical repeats leads to correct p-values and correct type I error control!
- What to do when you have an unequal number of technical repeats: more advanced methods are required
 - e.g. mixed models
 - The mixed model framework can model the correlation structure in the data
- Mixed models can also be used when inference between and within blocks is needed, e.g. split-plot designs.
- But, mixed models are beyond the scope of the lecture series.

Nature methods: Split-plot designs

[Nature Methods - Point of Significance - Split-plot Designs](#)